

SUMMARY OF THE SCIENTIFIC BACKGROUND

Experienced neuroscientist with 10+ years of expertise in behavioral systems neuroscience. Specialized in a range of rodent behavioral assays, neural circuits genetic manipulations, and neural activity recordings (both *in vivo* freely moving and head-fixed electrophysiology). Skilled in experimental setup design, optimization, and troubleshooting, as well as performing microsurgeries. Demonstrated success in securing fellowships, admissions to competitive training programs and travel grants. Proven leadership in mentoring Master's and Bachelor's students. Actively involved in organizing symposia and conferences, and in science popularization. Engaged member of the international community of young neuroscientists.

EDUCATION

- 01/01/2018 – 31/08/2023 **PhD in Biology** at Arenberg Doctoral School, KU Leuven Department of Science, Field of Biology
Dissertation: *Unraveling the cell type specific underpinnings that shape collicular mediated visual behaviors*
Supervisor: Dr. Karl Farrow (Neuro-Electronics Research Flanders NERF, Leuven, Belgium)
- 01/10/2015 – 21/11/2017 **Master's Degree in Neurobiology** at Jagiellonian University in Cracow, Poland
Dissertation: *Circadian analysis of the orexin fibers distribution in the rat lateral geniculate nucleus of the thalamus*
Supervisor: Prof. Dr. Marian H. Lewandowski (Department of Neurophysiology and Chronobiology, Institute of Zoology and Biomedical Research, Jagiellonian University in Cracow, Poland)
- 01/10/2012 – 08/07/2015 **Bachelor's Degree in Neurobiology** at Jagiellonian University in Cracow, Poland
Dissertation: *Role of GABA-ergic innervation in the mammalian biological clock*
Supervisor: Prof. Dr. Marian H. Lewandowski (Department of Neurophysiology and Chronobiology, Institute of Zoology and Biomedical Research, Jagiellonian University in Cracow, Poland)

RESEARCH EXPERIENCE

- 15/01/2024 - present **Postdoctoral research** at Bassem Hassan lab, ICM – Paris Brain Institute, Paris, France
Aim: To investigate links between brain wiring structure and physiology and their significance for behavior.
 - Neurodevelopmental fly genetics
 - Behavioral experiments
 - *In vivo* two-photon calcium imaging on behaving flies
- 10/03/2025 – 14/03/2025 **Visiting scientists** at Marion Silies Lab, Johannes-Gutenberg-University of Mainz, Mainz, Germany
Aim: Data analysis of calcium imaging signals
- 01/01/2018 – 31/08/2023 **PhD research** at Karl Farrow lab, NERF – empowered by imec, KU Leuven and VIB, Leuven, Belgium
Aim: To investigate the role of distinct neuronal cell types in visually-guided innate behaviors.
 - Open-field behavioral paradigms to study innate behaviors in mice
 - *In vivo* freely-moving and head-fixed Neuropixels recordings in mice
 - Cell-type specific optogenetics and chemogenetics
 - Viral injections
- 2012 October – 2017 November **Undergraduate research** under supervision of Dr. Marian H. Lewandowski at Jagiellonian University, Cracow, Poland
Aim: To investigate circadian changes of orexinergic innervation in rats' visual system.
 - Brain histology
 - Assistance for *in vitro* patch clamp recordings

2017 March – July	Internship at Karl Farrow lab, NERF, Leuven, Belgium <u>Aim:</u> To establish a behavioral protocol for visually-triggered orienting and defensive behaviors in a virtual reality environment. • Virtual reality environment for mice • Head-fixed behavioral protocols
October 2016 – February 2017	Internship at Behavioral Neurodynamics Lab (Ponomarenko/Korotkova Lab), Leibniz-Institute for Molecular Pharmacology (FMP), Berlin, Germany <u>Aim:</u> To investigate sleep and memory regulation by forebrain circuits. • Projection- and cell-type specific optogenetic interrogation • Local field potential recordings <i>in vivo</i> • Freely-moving behavioral protocols
2016 September	Internship at Laboratory of Neurodegeneration (Kuźnicki Lab), International Institute of Molecular and Cell Biology, Warsaw, Poland <u>Aim:</u> To investigate the role of calcium-related proteins involved in Huntington’s disease. • Development of <i>in situ</i> proximity ligation assay (PLA) on brain slices
2015 January - March	Erasmus+ Traineeship Program at Laboratory of Cell Biology of the Synapse (Matteoli Lab), Neuroscience Institute, CNR, Milan, Italy <u>Aim:</u> To investigate the role and molecular mechanisms of Eps8, the actin regulatory protein in synapse plasticity. • Western blot analysis

AWARDS AND DISTINCTIONS

2025	Carnot Training Grant for short collaborative visits
2025	Marie Skłodowska-Curie Actions Postdoctoral Fellowship
2024	The French Foundation for Medical Research (FRM) Postdoctoral Fellowship
2023	Fyssen Foundation Postdoctoral Fellowship
2021	Participant of Transylvanian Experimental Neuroscience Summer School (TENSS) 2020/2021 – competitive selection
2021	Participant of The Cajal NeuroKit Course: Visual Reactive Programming – Bonsai, organized by The CAJAL Advanced Neuroscience Training Programme in partnership with NeuroGEARS
2018 - 2022	Research Foundation – Flanders (FWO) PhD Fellowship (success rate ~20%)
2018	Travel Grant to attend The CAJAL Advanced Neuroscience Training Programme : Behaviour of Neural Systems (competitive selection: 20 people accepted, 4 awarded with travel grant), awarded by the CAJAL Programme, Champalimaud Centre for the Unknown, Lisbon, Portugal
2017	Travel Grant to attend 7th Neuron Technology Summer School: From Electrophysiology to Imaging and BioNanotechnology, awarded by International School for Advanced Studies (SISSA), Trieste, Italy
2015/2016 – 2016/2017	The Deans' Award for Outstanding Academic Achievement
2016	Laureate of the competition “Grasz o staż” (“Win the internship”)
2015	Organizing Committee Award for the Best Theoretical Poster – 5th International Conference Aspects of Neuroscience” (poster titled: <i>How does your brain measure a year? The role of GABAergic transmission in seasonal encoding</i>)
2014	Scientific Committee Award for the Best Theoretical Poster – 4th International Conference Aspects of Neuroscience (poster titled: <i>Exciting GABA! Depolarizing effects of the neurotransmitter</i>)

PUBLICATIONS

- *Rethinking naturalistic behavior: The Terzolas Manifesto*; **Anna Chrzanowska**, Mateusz Kostecki, Natalia Krasilshchikova, Matilde Perrino, Luigi Petrucco (all authors contributed equally) (The Open Science Framework 2025; doi: https://doi.org/10.31219/osf.io/ch3an_v2)

Impact: This work criticizes the usefulness of the concept of “naturalistic” in behavioral neuroscience.

Contribution: Conceptualization, manuscript writing and editing

- *Multiplexed Surface Electrode Arrays Based on Metal Oxide Thin-Film Electronics for High-Resolution Cortical Mapping*; Horacio Londoño-Ramírez, Xiaohua Huang, Jordi Cools, **Anna Chrzanowska**, Clément Brunner, Marco Ballini, Luis Hoffman, Soeren Steudel, Cédric Rolin, Carolina Mora Lopez, Jan Genoe, Sebastian Haesler. (Advanced Science 2023; doi: <https://doi.org/10.1002/advs.202308507>)

Impact: This work proposes an approach that solves a major problem in high-density brain-monitoring systems: it increases recording electrode count without additional wiring, by integrating flexible electronics which allow for signal multiplexing, bridging the gap between academic devices and clinical applications.

Contributions: *In-vivo* electrophysiology experiments, manuscript editing

- *Optogenetic fUSI for brain-wide mapping of neural activity mediating collicular-dependent behaviors*; Arnau Sans-Dublanc*, **Anna Chrzanowska***, Katja Reinhard, Dani Lemmon, Bram Nuttin, Théo Lambert, Gabriel Montaldo, Alan Urban# and Karl Farrow# (Neuron 2021; doi: 10.1016/j.neuron.2021.04.008) (*equal contribution)

Impact: In this work we demonstrated that optogenetic activation of different neuronal populations in the superior colliculus each triggers diverse behaviors, ranging. We combined functional ultrasound imaging with optogenetics to reveal the network of 264 brain regions functionally activated by these collicular cell types. Stimulating each neuronal group activated distinct sets of brain nuclei, including areas previously not thought to be part of collicular circuitry. This work offers insights into the functional organization of networks governing behaviors, and a method for exploring whole-brain activity downstream of targeted cells in small mammal

Contribution: Co-lead author, performed the key experiments, conceptualization of the study, experimental design, data analysis, manuscript writing and editing

- *Research culture: science from bench to society*; Lorenzo Canti, **Anna Chrzanowska**, M. Giulia Doglio, Lia Martina, Tim Van Den Bossche (Biology Open 2021; doi: 10.1242/bio.058919) (all authors contributed equally)

Impact: This article disseminates the values of research culture in science practices.

Contribution: Conceptualization, manuscript writing and editing

- *Multiple excitatory actions of orexins upon thalamo-cortical neurons in dorsal lateral geniculate nucleus – implications for vision modulation by arousal*; Lukasz Chrobok, Katarzyna Palus, **Anna Chrzanowska**, Mariusz Kępczyński and Marian Lewandowski (Scientific Reports 2017; doi:10.1038/s41598-017-08202-8)

Impact: This work offers insights into the impact of orexinergic modulation in the visual thalamus.

Contribution: Performed immunohistochemical studies, data analysis and interpretation, manuscript editing

- *Disinhibition of the intergeniculate leaflet network in the WAG/Rij rat model of absence epilepsy*; Lukasz Chrobok, Katarzyna Palus, Jagoda S Jeczmiern-Lazur, **Anna Chrzanowska**, Mariusz Kępczyński and Marian Lewandowski (Experimental Neurology 2017; doi: 10.1016/j.expneurol.2016.12.014)

Impact: This work shows for the first time the disinhibition of intergeniculate leaflet (IGL) neurons in a model of absence epilepsy, proposing the possible involvement of circadian-related brain structures in the epileptic phenotype.

Contribution: Performed immunohistochemical studies, data analysis and interpretation, manuscript editing

In preparation:

- *The role of collicular cell types in behavioral reactions to different threat intensities in mice*; Bram Nuttin* **Anna Chrzanowska***, Amber Bruynsteen, Arnau Sans-Dublanc# and Karl Farrow# (*equal contribution)

Impact: In this work we show how changes in the saliency of different danger-mimicking visual stimuli influence behavior and the role of different cell types in these responses. We demonstrate that behavioral reaction strength to these stimuli increases with the perceived threat intensity, with two distinct collicular cell types differentially modulating these responses.

Contribution: Co-lead author, performed the key experiments, conceptualization of the study, experimental design, data analysis, manuscript writing and editing

CONFERENCE PRESENTATIONS

Invited speaker:

- Chrzanowska A., Czajewski I. How to write a successful MSCA proposal; PORT for Health Neuroscience, Wroclaw, Poland, 9 Apr 2025
- Chrzanowska A. Cell vibes: how collicular cell-types contribute to guiding mice innate behaviors; Neurons in Action, Warsaw, Poland, 30 May 2025

Oral presentations:

- Chrzanowska A. *Towards embracing brain complexity: lessons from cell types of the superior colliculus*; The European Neuroscience Conference by Doctoral Students (ENCODS) Paris, France, 8 Jul 2022.
- Chrzanowska A., Sans-Dublanc A, Reinhard K, Lemmon D, Nuttin B, Lambert T, Montaldo G, Urban A, Farrow K. (2020). *A cell-type specific understanding of innate defensive behaviors: the collicular perspective*. Neuromatch 3.0 Conference, Virtual Conference, 26-30 Oct 2020.
- Chrzanowska A. *Using virtual reality to trigger visually guided innate behaviors*. NERF seminar, NERF, Leuven, Belgium, 28 Mar 2019.
- Chrzanowska A. *How can we use cell types to study brain and behavior?*. Joint Neuroscience seminar CBD-NERF-CMN, VIB-KU Leuven Center for Brain & Disease Research, Leuven, Belgium. 22 Oct 2022.

Poster presentations:

- Chrzanowska A., Bengochea M., Hassan B. *Neural circuits of individuality in the context of learning*. VIP & DIF 2024, Paris, France. 17 Jun 2024.
- Nuttin B, Kuhn N, Chrzanowska A., Sans Dublanc A, Farrow K. *Locomotion shapes visually-evoked neural dynamics through modulation of collicular populations*. Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal. 29 Feb 2024.
- Londono Ramirez H, Huang X, Cools J, Chrzanowska A., Coulson P, Léa F, Brunner C, Ballini M, Van Helleputte N, Mora Lopez C, Genoe J, Haesler S. *Actively Multiplexed μ ECoG Array Based on Thin-Film Electronics for High-Resolution Brain Mapping*. 11th International IEEE EMBS Conference on Neural Engineering (NER), Baltimore, United States of America. 25-27 Apr 2023.
- Nuttin B, Kühn N, Chrzanowska A., Sans Dublanc A, Farrow K. *Locomotion shapes visually-evoked neural dynamics through modulation of collicular subpopulations*. Bernstein Conference, Berlin, Germany. 26-29 Sep 2023.
- Chrzanowska A., Nuttin B, Grillet M, Sans Dublanc A, Gomez V, Ouriaghli Taouil C, Montaldo G, Urban A, Farrow K. *Coordinated activity of two cell-type specific circuits and their role in guiding behaviors – insights from the mice superior colliculus*. FENS Forum 2022, Paris, France. 11 Jul 2022.
- Nuttin B, Chrzanowska A., Sans Dublanc A, Gomez V, Kühn N, Farrow K. *Cell-type-specific contributions to visuomotor transformations in the superior colliculus*. FENS Forum 2022, Paris, France. 11 Jul 2022.
- Chrzanowska A., Nuttin B, Grillet M, Sans Dublanc A, Gomez V, Ouriaghli Taouil C, Montaldo G, Urban A, Farrow K. *Using different neuronal cell types to study the brain and behavior*. VIB CBD - NERF PhD Symposium, Leuven, Belgium. 27-28 Oct 2022.
- Chrzanowska A., Nuttin B, Sans Dublanc A, Farrow K. *Neural basis of behavioral habituation to visual threat*. VIB CBD - NERF PhD Symposium, Leuven, Belgium. 15 Oct 2021.

- Sans-Dublanc A, Chrzanowska A, Montaldo G, Reinhard K, Urban A, Farrow K. *Mapping cell-type functional projections of the superior colliculus using opto-fUSI reveals distinct downstream networks*. Bernstein Conference on Computational Neuroscience, Berlin, Germany, 17-20 Sep 2019.
- Chrzanowska A, Avelar C, Moustakis I, Sans-Dublanc A, Farrow K. *Using virtual reality to reliably trigger visually guided behavior in head fixed animals*. FENS Forum of Neuroscience 2018, Berlin, Germany, 07-11 Jul 2018.

STUDENT SUPERVISION

Supervision of bachelor students' projects:

- Sofia Rebiha Decesaro:
 - *Comparative transcriptomic profiling of Drosophila melanogaster brains in contrasting object fixation behavioral phenotypes*.
 - Teaching of behavioral experiments in Buridan setup, fly handling, data analysis
 - Summer intern in September 2024
- Diana Dayekh:
 - *Advancing the understanding of Drosophila visual processing through experimental design and integration of aversive conditioning paradigm into a classic Buridan setup*.
 - Teaching of behavioral experiments in Buridan setup, fly handling, supervising design and conceptualization of modified version of Buridan setup to perform aversive conditioning with electric shocks, data analysis
 - Academic year: 2023 – 2024
- Martyna Palys:
 - *Role of narrow-field cell types of the mice superior colliculus in triggering innate defensive behaviors - optogenetic approach*.
 - Teaching of optogenetics experiments in freely-moving mice, microsurgeries, brain histology, data analysis
 - Academic year 2022 – 2023.
- Sade Adeyemi:
 - *Evolutional differences of socially mediated response to visual threats*.
 - Teaching of social experiments in laboratory (*Mus musculus*) and deer mice (*Peromyscus maniculatus* and *Peromyscus polionotus*), data analysis
 - Academic year 2022 – 2023.
- Diethe Peeters, Tom Voet and Mathis Van den Bruel:
 - *Habituation of mice defensive behaviors triggered by predator-mimicking visual stimuli*.
 - Teaching of visually-guided behavioral experiments in an open field setup in mice, data analysis
 - Academic year 2021-2022.
- Mathilda Corcoles:
 - *Cell-type specific basis of defensive behaviors*.
 - Teaching of optogenetics experiments in freely-moving mice, microsurgeries, brain histology, data analysis

- Academic year 2020-2021.

Supervision of master students' projects:

- Helin Polat
 - *Genetic control of the timing of synaptogenesis*
 - Fly genetics
 - Academic year 2024-2025
- Hala Rachel El Hajj
 - *Neural circuits of individuality in the context of social influence*
 - Teaching of behavioral experiments in the Buridan setup; fly handling
 - Academic year 2024-2025
- Chaimae Ouriaghli Taouil:
 - *The superior colliculus cell-types and their role in controlling mouse defensive behaviors.*
 - Teaching of visually-guided behavioral experiments in an open field setup in mice, virus injections and microsurgeries, brain histology, data analysis
 - Academic year 2021-2022.
- Liuyin Yang and Nidaa Elsayed:
 - *Clock synchronization for correlation of behavioral and neural measurements in mice VR experiments.*
 - Supervising the conceptualization of setup design
 - Academic year 2019-2020.
- Carolina Tigre Avelar:
 - *Developing a Virtual Reality environment to study mice behavior.*
 - Supervising the conceptualization of setup design
 - Academic year 2017-2018.

MAJOR COLLABORATIONS

- **Alan Urban, NERF – empowered by imec, KU Leuven and VIB.** Together with the team of Alan Urban and Karl Farrow (both at NERF), we established a novel technology to study cell-type specific brain-wide circuits in mice, called optogenetic functional ultrasound imaging (opto-fUSI). This project was published in a high impact journal during my PhD. Alan Urban's team is developing fUSI for basic and clinical neuroscience research.
- **Nanoengineering team of imec (Interuniversity Microelectronics Centre, Leuven, Belgium).** I have collaborated with the team of neuroengineers at imec to test *in vivo* a novel device for large-scale brain recordings. The outcome of this collaboration was successfully published.

CONTRIBUTION TO SCIENTIFIC COMMUNITY

Teaching:

- Teaching assistant at Transatlantic Behavioral Neuroscience Summer School (3D printing, Bonsai reactive programming, Arduino and electronics), 2021 – present
- Local teaching assistant at NERF for the Cajal NeuroKit Course: Visual Reactive Programming – Bonsai, 24-28/10/2022

